MANISH HIRA

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Research Interests

Artificial Intelligence and Machine Learning, Sufficient Dimension Reduction, Causal Inference, High-Dimensional Modeling, Marketing Analytics, Credit Risk Modeling, Financial Engineering, Computational Finance, Optimization, Statistical Inference, Time Series Analysis

Education

Virginia Commonwealth University, Richmond, VA
Ph.D. Candidate, Systems Modeling and Analysis 2024 – Present

- Joint program between Department of Statistical Sciences and Operations Research and Department of Mathematics and Applied Mathematics
- Focus: Development of mathematical and computational skills for conceptualizing and analyzing real-world systems
- Coursework: Mathematical Statistics, Linear Modeling, Research Exploration, Graph Theory, Optimization
- Working Paper: Sufficient dimension reduction; aligning research to causal inference and high-dimensional modeling relevant to marketing analytics and credit risk

University of Washington, Seattle, WA
M.S., Computational Finance & Risk Management 2019 – 2021 GPA: 3.7

- Coursework: Investment Science, Financial Data Science, Asset Allocation & Portfolio Management, Options and Derivatives, Monte Carlo Methods in Finance, Optimization Methods in Finance, Risk in Financial Institutions, Credit Risk Management, Machine Learning for Finance, Partial Differential Equations
- Distinguished Projects:
  - Mortgage Default Credit Risk Model using Machine Learning (Fannie Mae dataset; algorithms: Decision Tree, Random Forest, XGBoost, Logistic Regression, KNN, DNN, CNN)
  - Retirement Portfolio with Market Forecast (recursive dynamic programming for 6 client profiles with portfolio optimization)

University of Notre Dame, Mendoza College of Business, Notre Dame, IN
MBA, Consulting & Finance (Graduated Cum Laude) 2013 – 2015 GPA: 3.6

- GMAT: 750
- Coursework: Statistical Inference, Forecasting & Predictive Analytics, Market Research, Corporate Finance, Economics, Investments, Equity Analysis, Fixed Income Process, Mergers and Acquisitions, Global Investment Management

University of Delhi, Netaji Subhas Institute of Technology (NSIT), Delhi, India
B.E., Instrumentation & Control Engineering 1996 – 2000

- Consultant (intern) for Center for Development of Telematics (government-funded R&D project: Linux-based networking application using socket programming)

Professional Experience

Capital One, Richmond, VA
Data Analysis Manager — January 2019 – Present

- Led analytical projects for brand and marketing to develop insights informing strategic decision-making
- Evaluated advertising experiments for product launches in 10 markets, informing $50M media allocation
- Impacted executive decision to allocate baseline brand marketing spend of 20% of total budget using 20-year media investment analysis
- Formulated architectural blueprint and prototype for an enterprise **Agentic Brand Marketing Analytics Stack**, deploying a hierarchical multi-agent swarm over a dynamic 5D signal cube foundation (*Channel* × *Touchpoint* × *Segment* × *Creative* × *Time*) governed by an enterprise semantic ontology (SOV, SOW, NAB) and strict Human-in-the-Loop (HITL) compliance guardrails
- Engineered multi-modal ingestion pipeline converting unstructured creative assets (video, copy, landing pages via OCR/vision tagging) and 1st-party portfolio signals into first-class inputs for automated downstream econometric and MMM analysis
- Developed "State of the Business" newsletter for executive audience; guided analytical integrity and presented findings to senior leadership audiences
- Managed team of analysts and data engineers; extensive talent and performance management
- Utilized Databricks in AWS cloud for large datasets; built quantitative solutions and dashboards in Tableau
- Attended conferences (AudienceXScience) for thought leadership and knowledge sharing
- Spearheaded AI tooling adoption across team: deployed custom Gemini Gems for standardized tech templates and data governance enforcement, and introduced GitHub Copilot and Claude Code as agentic coding assistants, significantly improving team delivery speed and code quality
- Conducted purchase volume (PVOL) geographic market analysis that directly shaped media targeting strategy; accelerated delivery of time-sensitive geographic analysis (weeks-long effort turned around in days) to meet escalated business need
- Led data governance initiative migrating unstructured vendor data from siloed spreadsheets to Snowflake, eliminating data silos, mitigating compliance risk, and securing project deliverables on schedule
- Resolved big-data processing bottlenecks for high-volume consumer clickstream datasets; provided architectural guidance that unblocked downstream analytics pipeline
- Participated in Program Increment (PI) planning to align brand technology roadmap with brand intelligence top priorities, incorporating self-service tooling, ETL modernization, and agentic AI capabilities

State of Virginia / Capital One, Richmond, VA
Data Analytics Consultant — November 2015 – December 2018

- Data and analytics consulting assignments for financial services and retail clients
- Delivered targeted card marketing campaigns; segmentation and targeting for 50M+ customers and $500M quarterly transactions
- Built machine learning models using credit card transaction data to predict first-time shoppers at retail partners (XGBoost; AUC: 0.81)
- Designed and developed cloud-based ETL/ELT pipelines for retail operations

- Data engineering from source to staging to enterprise data warehouse using Unix scripting and Oracle

Samsung Electronics, South Korea & India
Senior Manager — January 2009 – August 2013

- Planned and managed product portfolio of Samsung mobile products impacting sales of ∼100M units
- Founded Intellectual Property Cell for Innovation Management, generating 18 granted patents and institutional IP pipeline
- Developed Business Intelligence reporting dashboards using MS SQL Server, SSIS/SSRS
- Mentored team of engineers to develop embedded software for ultra-low-cost mobile devices

Wipro Technologies, USA, Finland, Japan, India
Technical Lead — September 2000 – January 2009

- Developed multimedia projects worth $10M+ for Qualcomm, resolving OEM commercialization issues for MSM7x00 chips
- Implemented automated mobile test framework for Nokia Mobile in Finland (Linux-based open-source software on TI OMAP platform)
- Coordinated development of MRI scanning software according to DICOM standards ($2M project for Toshiba)

Publications

Patents

1. Hira, M. (2023, 2024). Smart Card Authentication System. *United States Patent Nos. US11836733B2 & US12154116B2*. Capital One Services, LLC.
2. Hira, M., & Agrawal, S. K. (2015). Near Field Communication Method and Apparatus Thereof. *United States Patent No. US9160418B2*. Samsung Electronics Co., Ltd.
3. Hira, M., & Agrawal, S. K. (2015). A Method And System For Secured Data Transmission During Near Field Communication (NFC) Attack. *India Patent No. IN954DE2012*. Samsung Electronics Co., Ltd.

Journal Articles

4. Hira, M., & Koppuravuri, V. H. (2025). Volatility Modeling Using GARCH and Machine Learning Hybrids. *Well Testing Journal*, 34(S3), 417–435.
5. Hira, M., & Madhubabu, K. (2025). A Literature Review on the Role of Machine Learning, Deep Learning and AI in Real-Time Ad Personalization and Sentiment Analysis on Social Media Platforms. *Journal of Artificial Intelligence and Research Applications*, 5(1), 1–31.
6. Koppuravuri, V. H., Hira, M., Budige, K., & Kumar, A. (2026). Authenticity-Creativity Tradeoffs in Generative AI Representations of Cultural Heritage: Voice AI Integration in Music Reconstruction. *Scientific Culture*, 12(5), 2719–2733. DOI: 10.5281/zenodo.21736803.

Conference Proceedings

7. Mandal, G., Baladithya, D., Likhita, D., Goud, M. S., Kumar, M. V., & Hira, M. (2026). Integrating Deep Learning and Traditional Techniques for Underwater Image Enhancement. In

Proceedings of the 5th International Conference on Artificial Intelligence and Signal Processing (AISP 2025).

Working Papers

8. Hira, M. (In progress). Sufficient Dimension Reduction for High-Dimensional Marketing Analytics and Credit Risk Modeling. Ph.D. dissertation, Virginia Commonwealth University.

Citation Metrics

Metric	All Time	Since 2021
Citations*	7	4
h-index	1	1
i10-index	0	0

*Citations aggregated across Google Scholar (1 all-time/1 since 2021), Google Patents citing-patent records for US9160418B2 (4 all-time/1 since 2021), and Google Patents citing-patent records for US12154116B2 (2 all-time/2 since 2021), as of August 2026.

Professional Certifications

- **AI/BI for Data Analysts** | Databricks Academy — July 2026
- **Explainable AI (XAI) Specialization** | Duke University (Coursera) — April 2026
- **PyTorch for Deep Learning** | DeepLearning.AI (Coursera) — April 2026
- **QBronze (Quantum Computing and Programming)** | The Washington Institute for STEM, Entrepreneurship and Research, 2025
- **AWS Machine Learning – Specialty** | Amazon Web Services
- **AWS Data Analytics – Specialty** | Amazon Web Services (DynamoDB, Kinesis Analytics, Amazon Elasticsearch Service, Redshift, RDS, Aurora, Lambda, Glue ETL, QuickSight)
- **MLOps Specialization** | Coursera (Machine learning in production, data lifecycle, modeling pipelines, deployment)
- **Machine Learning Engineer Nanodegree** | Udacity (Supervised/Unsupervised Learning, Reinforcement Learning, SVMs, Neural Networks, Clustering, Dimensionality Reduction, Recommender Systems, Deep Learning, Anomaly Detection)
- **Machine Learning** | Stanford University (Coursera)
- **Quantitative Analysis in R** | DataCamp (Time Series Analysis, ARIMA, Forecasting, Portfolio Analysis, Credit Risk, Bond Valuation, Trading in R)

Technical Skills

Programming Languages: Python, SQL, PySpark, R, C/C++, Unix Shell, PyTorch, TensorFlow

Tools & Platforms: Snowflake, Databricks, Tableau, Looker, Power BI, Airflow, ODI, Teradata, Amazon Web Services (AWS), EMR, Glue, Lambda, SageMaker, Jupyter, Linux

Project Management: Agile (JIRA), Git/GitHub, Perforce, CVS

Machine Learning & Statistical Methods: Linear/Logistic/Lasso Regression, Decision Trees, Random Forest, XGBoost, Support Vector Machines, Clustering, Probabilistic Models (Gaussian,

Bayesian Neural Networks), A/B Testing, Window Functions, Query Performance Tuning

ML Libraries: Scikit-Learn, Pandas, Matplotlib, Seaborn, NumPy, SciPy, ggplot2, PySpark, Plotly, QuantLib

Professional Service & Leadership

Peer Review Activities

- Peer Reviewer, ICCSTE 2026 (International Conference on Computer Science and Technology Engineering), July 2026
- Peer Reviewer, ICIIBMS 2026 (International Conference on Intelligent Informatics and Biomedical Sciences), July 2026
- Technical Committee Member, IEEE SPML 2026 (9th International Conference on Signal Processing and Machine Learning), May 2026
- Peer Reviewer, IEEE iGET 2026 (Global Emerging Technology Conference), October 2026
- Peer Reviewer, IEEE AMLDS 2026 (Advanced Machine Learning & Data Science), July 2026
- Peer Reviewer, ICNCCom 2026 (International Conference on Networks, Computers and Communications), July–August 2026
- Peer Reviewer, IEEE TEMSMET 2026 (IEEE International Conference on Technology, Engineering, Management for Societal Impact), July–October 2026
- Peer Reviewer, NILES 2026 (8th Novel Intelligent and Leading Emerging Sciences Conference, Nile University, Egypt; IEEE), August 2026
- Peer Reviewer, EDIGE 2026 (International Conference on Emerging Digital Intelligence and Generative Engineering, VIT Bhopal University, India), August 2026
- Peer Reviewer, AIHC 2027 (International Conference on Artificial Intelligence in Healthcare, Symbiosis Institute of Technology, Hyderabad, India), January 2027
- Peer Reviewer, AIPE 2027 (4th International Conference on Artificial Intelligence and Power Engineering), October 2026
- Peer Reviewer, IEEE ICMACC 2026 (3rd International Conference on Recent Trends in Microelectronics, Automation, Computing and Communications Systems), August 2026
- Peer Reviewer, IEEE ICEC 2026 (2nd International Conference on Intelligent Computing and Emerging Communication Technologies), 2026
- Peer Reviewer, ICEECE 2026 (International Conference on Electrical, Electronics and Computer Engineering), 2026
- Peer Reviewer, ICon INDIA 2026 (International Conference on Intelligent Networks and Data-driven Intelligent Applications, Puducherry, India), September 2026
- Technical Auditor & Code Referee, ACGF (Authenticity-Constrained Generative Framework) Project: Conducted technical codebase quality audits, commit reviews, and research evaluation suite verification, 2026

Judging Activities

- Judge, RVAHack26 | Virtual Hackathon co-hosted by ACM Richmond, IEEE Richmond Section, IEEE TEMS, and IEEE Computer Society, October–November 2026
- Judge, CODiE Awards 2026 | Best AI Data & Analytics Solution (Developer & AI Tech category)
- Judge, VCU Graduate School 29th Annual Research Symposium, 2026
- Judge, AI Excellence Awards 2026 | Business Intelligence Group (bintelligence.us)
- Judge, Hack UNCP 2026 | University of North Carolina at Pembroke Hackathon

- Judge, CES Awards 2025 (AI-enabled Education Solutions category)
- Judge, Codie Awards 2025 (AI-enabled Education Solutions category)
- Judge, AI for Good Hackathon 2025
- Judge, VCU Graduate School 28th Annual Research Symposium, 2025

Professional Memberships

- IEEE Senior Member (Member #101554119) | 2025 – Present
- Association of National Advertisers (ANA) Committees | Committee Member (AI Committee, Analytics & Data Science, Brand Value & Measurement, MarTech), 2025 – Present
- INFORMS (Member #2006027)
- American Mathematical Society
- American Mensa

Volunteer Consulting

- Provided data, analytics, and technology strategy consulting to local small businesses through Taproot Foundation (e.g., Auto Glass Club via Capital One Pro Bono Marathon)
- Provided data and technology consulting services to local small businesses through Partnership for Housing Affordability (PHA)
- Provided data and technology consulting services to local small businesses through Startup Virginia

Training & Development

- Attended Capital One Leads training program
- Attended Professional Analyst Development Program (PADP), signature program for analyst community
- Committee Member, Breakouts & Lightning Talks Committee | Capital One Modeling & Analytics Conference (MAC) 2026 — Selected and curated breakout session topics and lightning talks for the internal annual analytics conference

Conferences & Presentations

- AudienceXScience Conference (Attendee; knowledge sharing with department on ongoing trends in marketing analytics)
- Conference Abstract Submitted, 2026 Conference on Artificial Intelligence, Machine Learning, and Business Analytics | Temple University Fox School of Business, Philadelphia, PA, December 10–12, 2026 — "Generative Engine Optimization in Enterprise Marketing: Multi-Stage Neural Dynamics, Competitive Equilibria, and Causal Attribution" (Under Review)
- Invited Speaker / Delivered Presentation, MSI/ARF Analytics & Forecasting Conference | Cornell Tech, New York City, NY, September 14–15, 2026 — "GEO in the Loop: Integrating Generative Engine Optimization into Marketing Analytics Workflows — Methodology, Metrics, and Governance"
- Invited Speaker, All Day AI 2026 Virtual Conference | BUILD Track (Virtual, Global Broadcast), October 22, 2026 — "GEO in the Loop: A Framework for Integrating Generative Engine Optimization Into Marketing Analytics Workflows"

- Featured Speaker, Richmond DA Connect (Associate TED Talks: Big Trends, Bold Accomplishments) | Capital One DA Executive Council, Richmond, VA (hybrid in-person, watch parties, and Zoom enterprise broadcast), September 22, 2026
- Paper Accepted – Not Presented, INFORMS Marketing Science Conference (ISMS 2026) | Carcavelos, Portugal, June 11–13, 2026 — Accepted as a regular (oral) presentation; unable to attend due to travel constraints

Honors & Awards

- Arthur Foundation Notre Dame MBA Fellow (100% tuition scholarship) | University of Notre Dame, 2013–2015
- GMAT Score: 750 | Graduate Management Admission Test
- Pi Mu Epsilon Inductee | Virginia Commonwealth University, March 2026 — National mathematics honor society; inducted in recognition of scholarly achievement in mathematics.